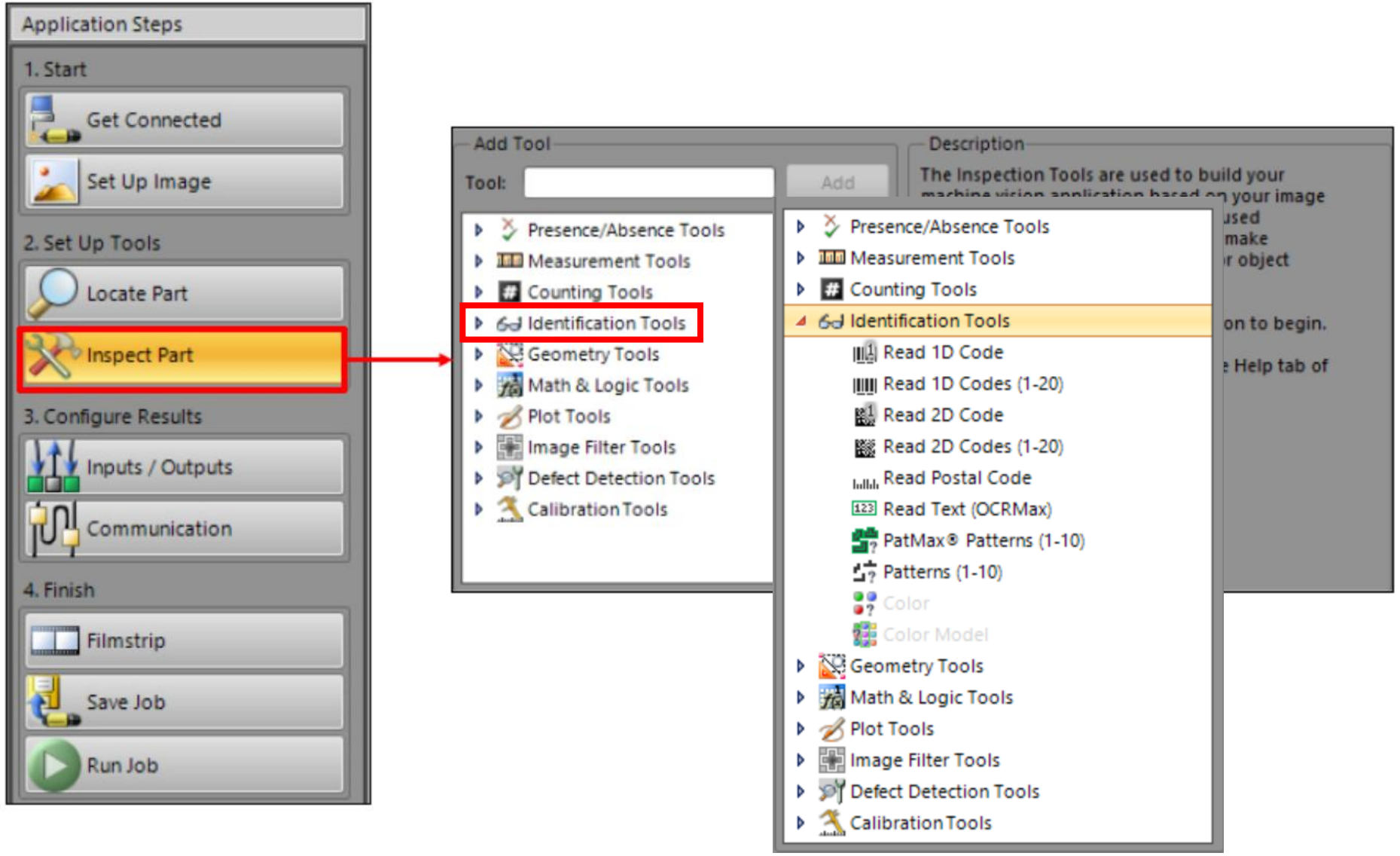
Identification and Image Filter Tools

Lecture 5

Identification Tools

Identification Tools

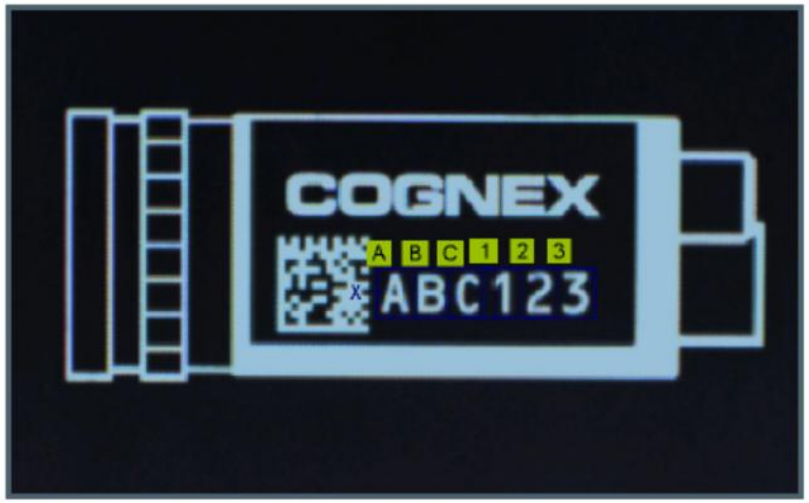


What Are Identification Tools?

- **Identification Tools** are used to identify and verify **1D and 2D barcodes, symbols, alphanumeric text, pattern features, and colors** in an image.



Read a code

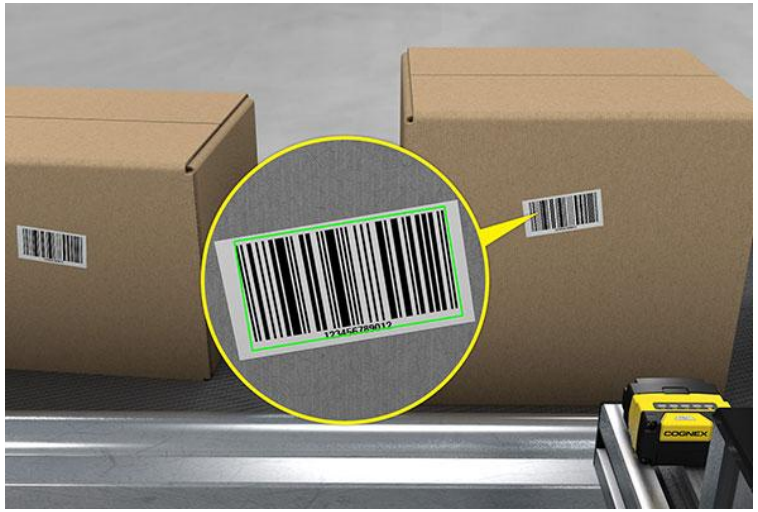


Read Text predefined or trained

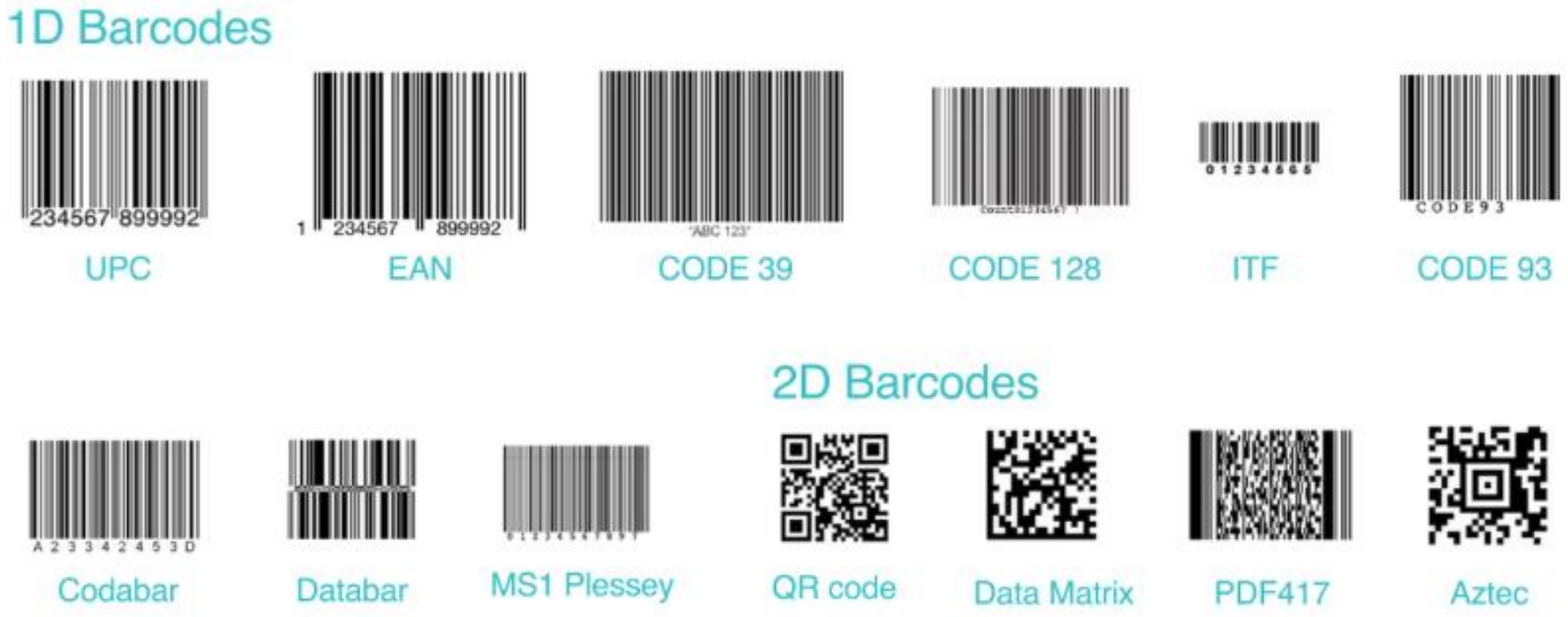
Barcode

- **Barcodes** are widely used in logistics, manufacturing, and retail as machine-readable optical labels that encode **alphanumeric data**.
- Their widespread use is due to their ability to enable **automated identification and data capture**, supporting **high-speed scanning, error reduction, and integration with database-driven inventory and tracking systems**.





1D & 2D Barcodes



- https://en.wikipedia.org/wiki/Barcode

1D & 2D Barcodes

1D barcodes (linear barcodes):

- Depending on the barcode symbology, 1D barcodes can contain **up to 85 alphanumeric characters**.
- These characters are encoded using **parallel bars and spaces**, where the widths of the bars and spaces represent specific characters.
- The encoded data typically represents a **product identifier**, which is linked to detailed product information stored in a database.



1D & 2D Barcodes



2D barcodes:

- **2D barcodes** store data in both **horizontal and vertical directions**, allowing them to encode significantly more information than 1D barcodes.
- Depending on the symbology, they can contain up to **3,116 numeric characters** or **2,335 alphanumeric characters**, encoded using **squares, dots, hexagons, or other geometric shapes**.
- **2D barcodes have built-in error correction**, allowing them to be read even when part of the code is damaged or obscured. For example, a **Data Matrix** code uses error-correction techniques to reconstruct missing or damaged data, increasing the likelihood of a successful read.

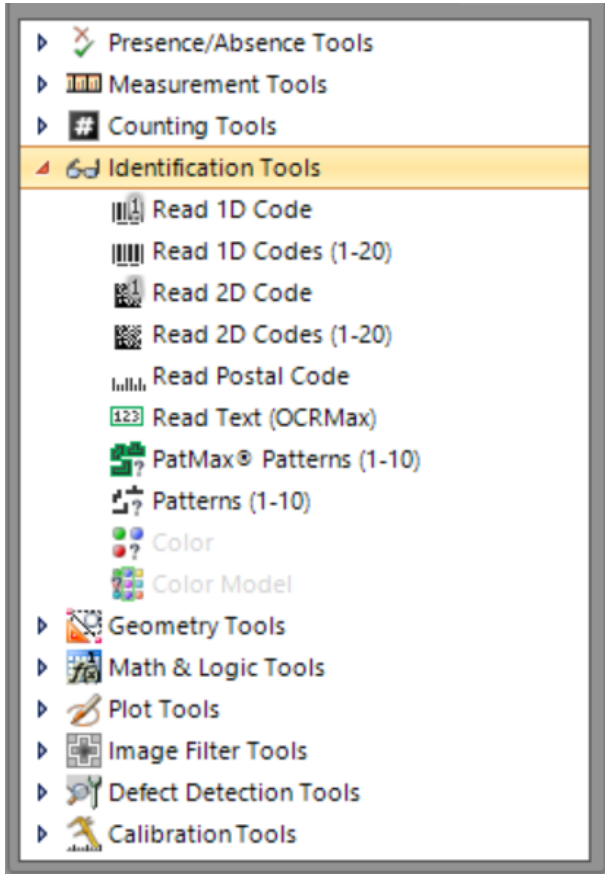
Data Matrix vs. QR Code



Feature	**Data Matrix**	**QR Code**
Shape	Usually square; can also be rectangular	Square
Finder pattern	**L-shaped border**	**Three large squares** in the corners
Data capacity	Generally smaller	Generally larger
Typical use	**Industrial/manufacturing** — part marking, traceability, electronics, medical devices	**Consumer/general-purpose** — websites, payments, tickets, contact information
Readability	Very small codes can be printed and read reliably	Usually requires a somewhat larger area
Scanning	Commonly used with **industrial vision systems**	Commonly scanned by **smartphones**

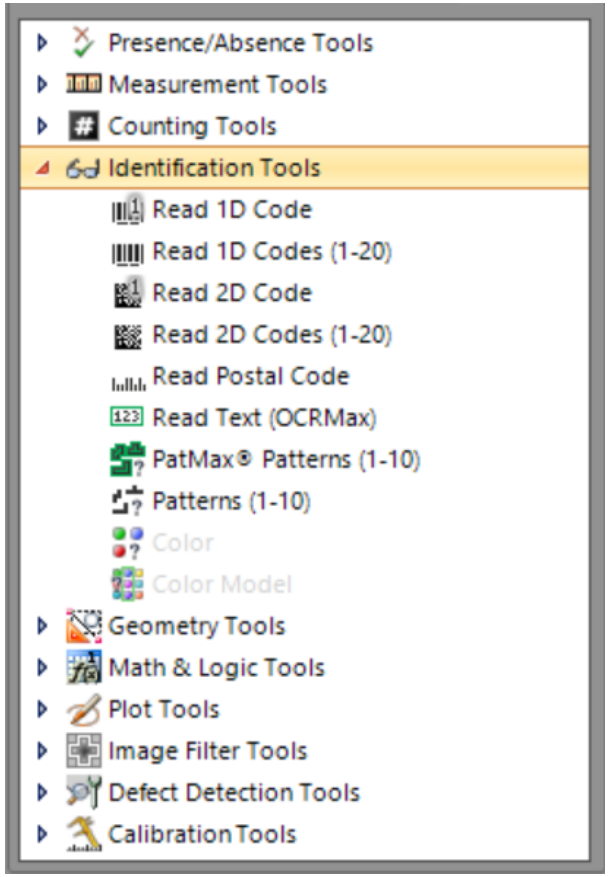
Identification Tools

- The **Read 1D Code** and **Read 1D Codes (1–20)** Tools decode and/or verify **1D barcodes** using the **ReadIDMax® algorithm**.
- The **Read 2D Code** and **Read 2D Codes (1–20)** Tools decode and/or verify **QR Code and Data Matrix** symbologies using the **ReadIDMax® algorithm**.
- The **Read Postal Code** Tool decodes and/or verifies a **single postal code** within a defined region using the **ReadIDMax® algorithm**.
- The **Read Text (OCR Max)** Tool reads and/or verifies a **text string** within a defined region after training and creating **user-defined character fonts**.

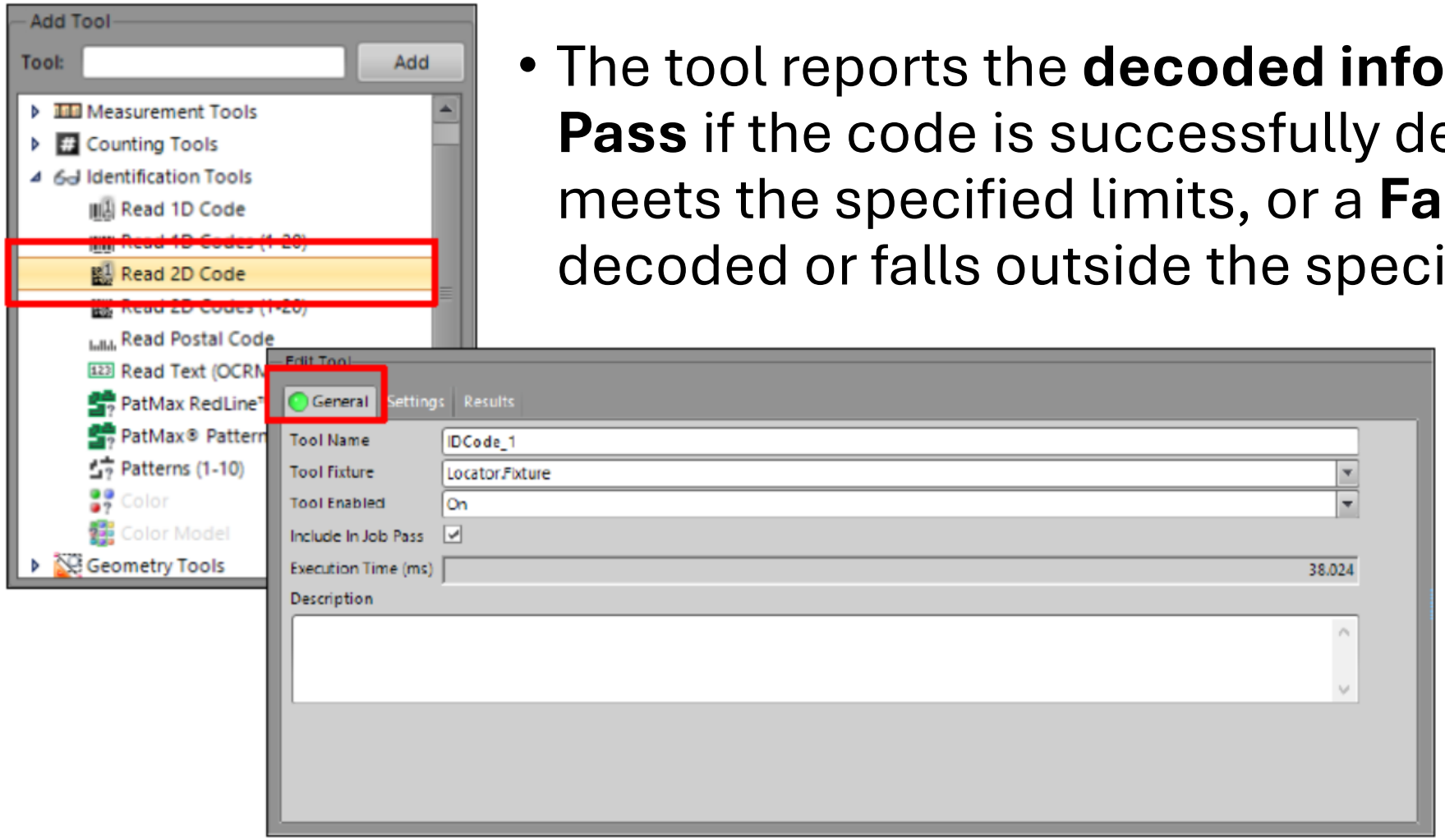


Identification Tools

- The **PatMax® Patterns (1–10)** or **Patterns** Tool is used to **train pattern features** using PatMax® algorithms and then **verify or identify** occurrences of those patterns in the image.
- The **Color** Tool determines, from a library of **trained colors**, which color best matches the color in the image.
- The **Color Model** Tool determines, from a library of **trained color models**, which color model best matches the colors in the image and returns the **name of the matching color model**.



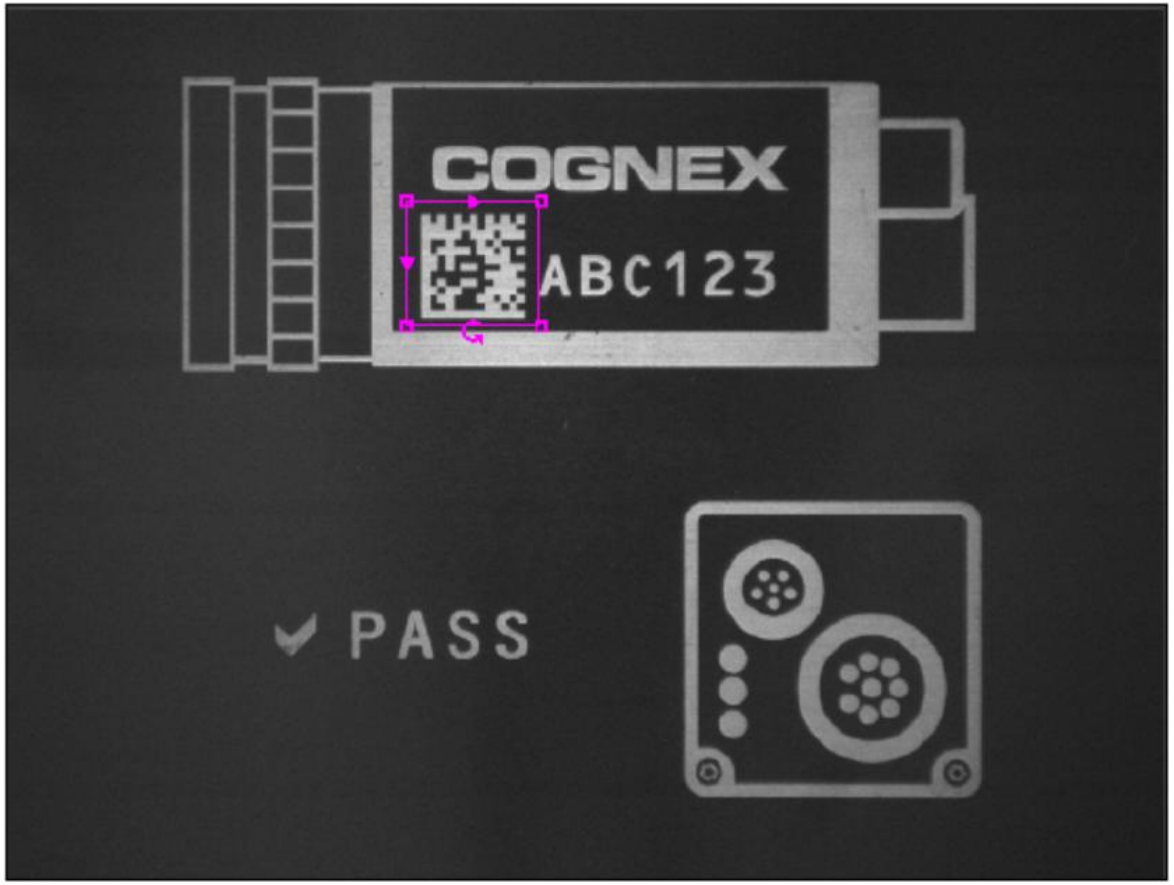
Read 2D Code Tool



- The tool reports the **decoded information** and a **Pass** if the code is successfully decoded and meets the specified limits, or a **Fail** if it cannot be decoded or falls outside the specified limits.

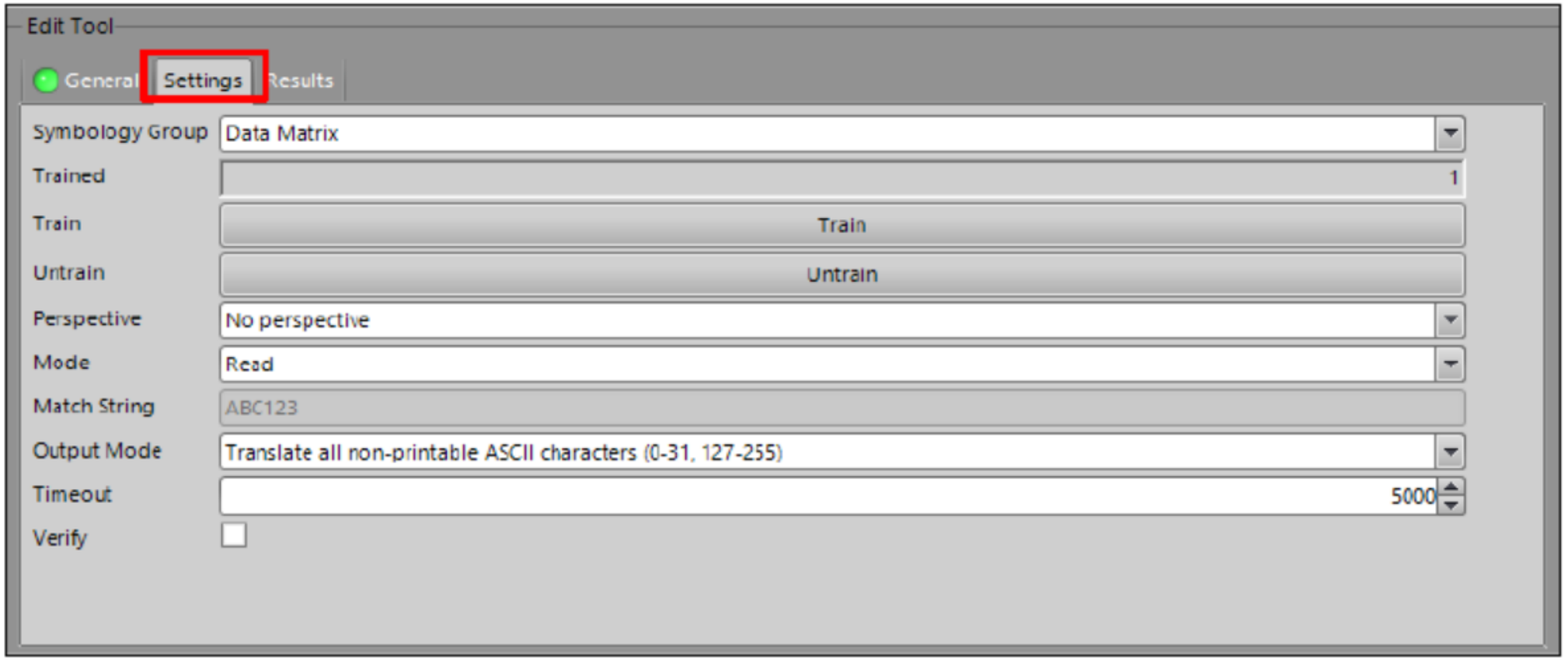
Read 2D Code Tool

- Position the region to the area of the 2D code.



Read 2D Code Tool

- Settings Tab



Read 2D Code Tool

The **Read 2D Code Tool's Settings Tab** contains the following information:

- **Symbology Group** – Defines the type of symbology (**Data Matrix or QR Code**) to decode. The default is **Data Matrix**.
- **Trained** – Indicates whether the tool has been trained on a particular code/symbol. It displays **1** when the tool has been trained and **0** when it has not.
- **Train** – Trains a model of the code/symbol. The trained information is retained until the **Untrain** button is pressed. Training can improve the **speed and performance** of the tool by teaching it which type of code/symbol to decode.

Read 2D Code Tool

- **Untrain –** Removes the training information from a trained code/symbol. To retrain the tool, press **Untrain**, then press **Train** with the new code present.
- **Perspective –** Defines the decoding setting (**No Perspective, Perspective, or Train Perspective**) for **Data Matrix** symbology. The **Symbology Group** must be set to **Data Matrix** to enable this parameter.
  - A **Data Matrix** code is ideally **square or rectangular**, with aligned rows and columns.
  - When the camera views the code at an **angle rather than perpendicular to its surface**, the code may appear **trapezoidal or skewed**. This distortion is called **perspective distortion**.
  - The **Perspective** setting tells the decoding algorithm whether to **account for and correct perspective distortion** when decoding the Data Matrix code.

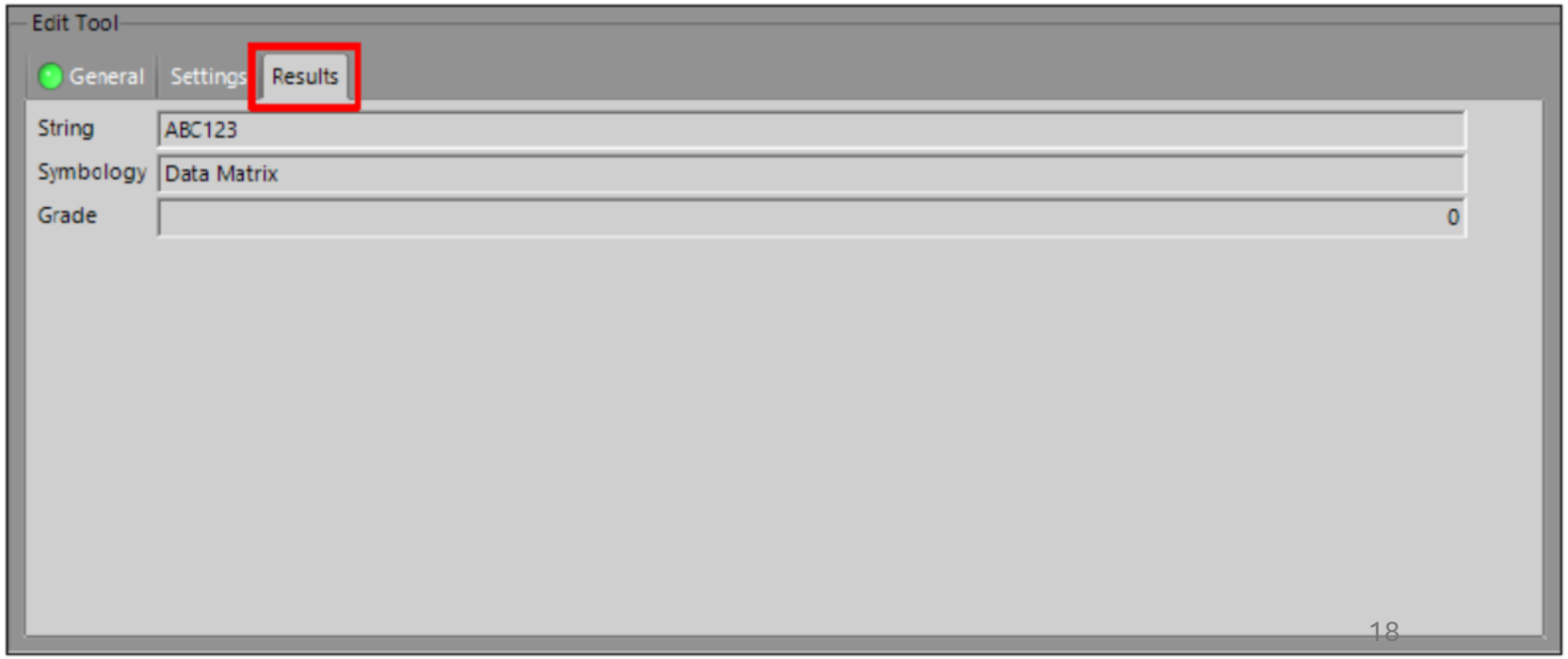
Read 2D Code Tool

- **Mode** – Defines the operational mode (**Read** or **Match String**; default = **Read**) of the tool.
- **Match String** – Defines the **expected string** to be decoded. This parameter is enabled only when **Mode** is set to **Match String**.
- **Output Mode** – Defines how the tool processes **non-printable ASCII characters**.
- **Timeout** – Defines the maximum time, in milliseconds (**0–30,000 ms; default = 0**), that the tool searches for a valid code/symbol before stopping and returning a **Fail**.
- **Verify** – Defines whether **print quality tests** are performed. This setting is **disabled by default**.

Read 2D Code Tool

The **Results** Tab displays the following information about the decoded code/symbol:

- **String** – Returns the **data string** contained in the decoded code/symbol.
- **Symbology** – Returns the **type of symbology** that was decoded.
- **Grade** – Returns the **quality grade** assigned to the decoded code/symbol.



Read 2D Code Tool





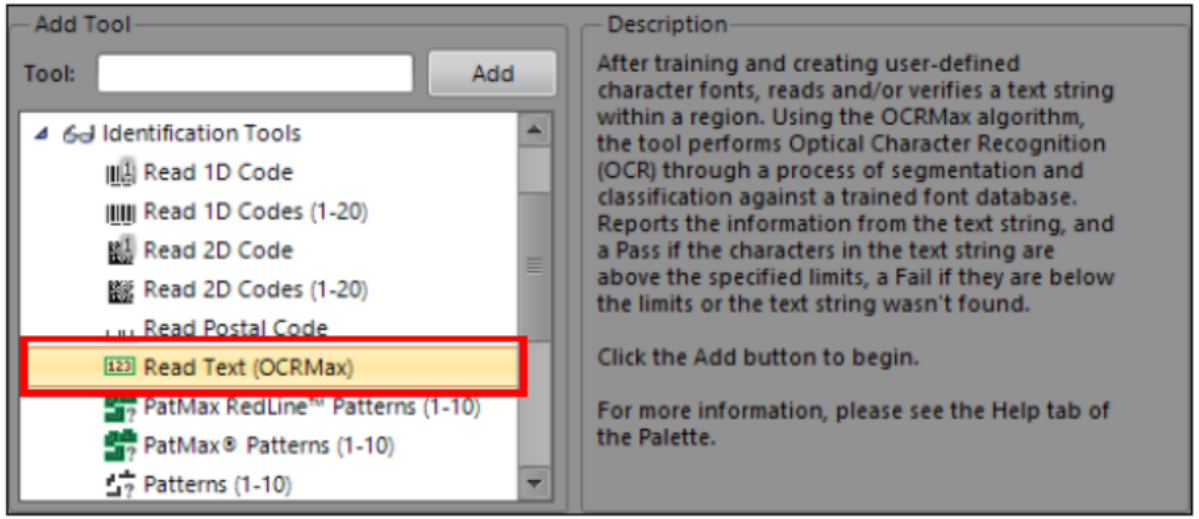
Mode set to *Read*

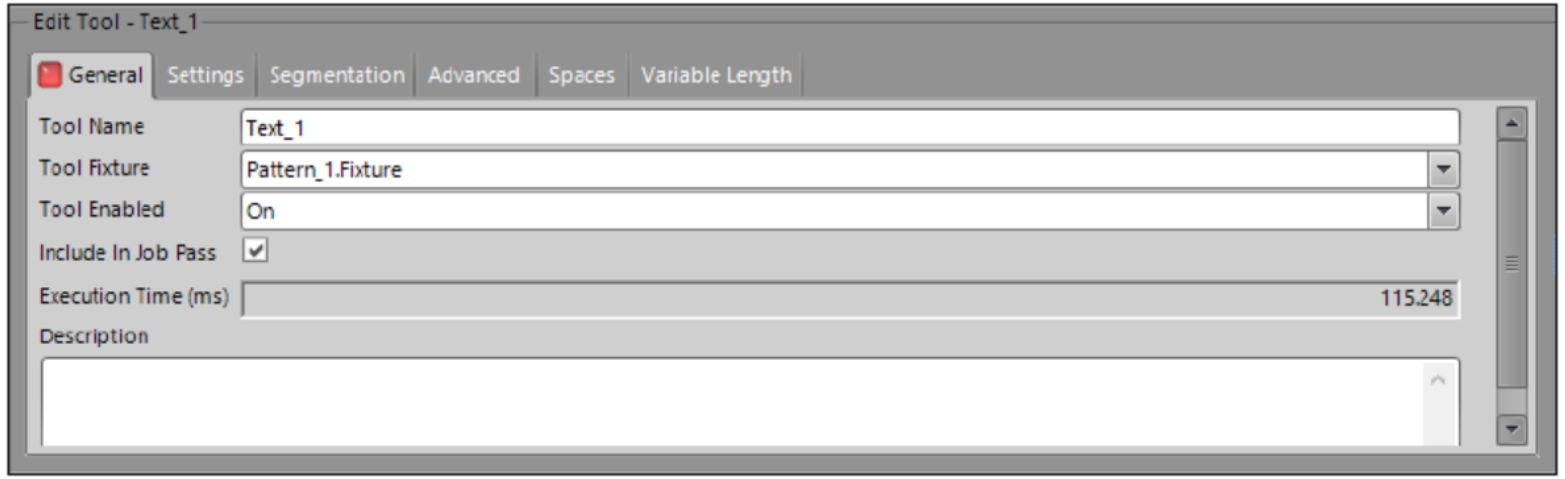




Mode set to *Match String*

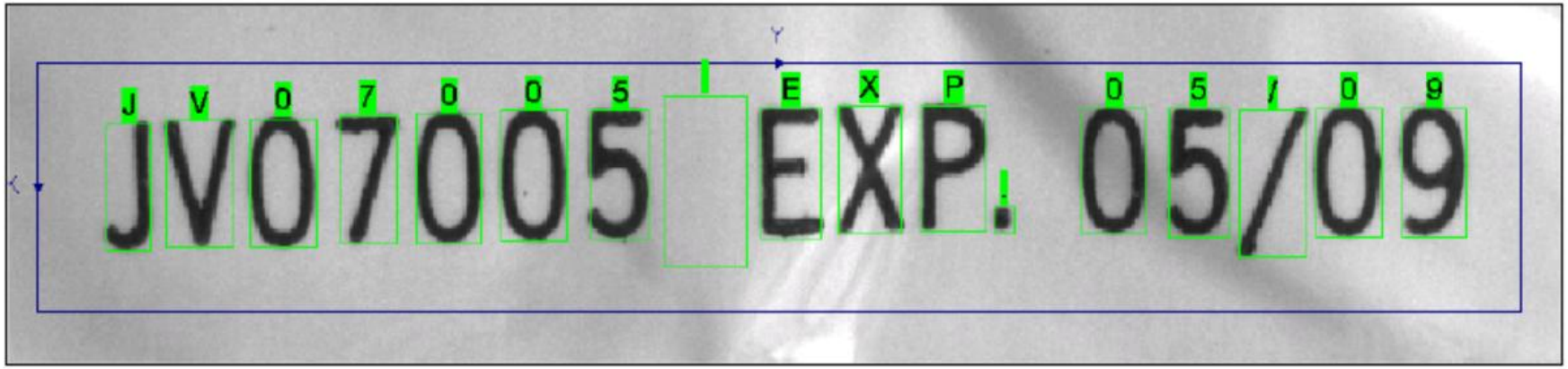
Read Text (OCRMax)





Read Text (OCRMax)

- The **Read Text** Identification Tool is used to **read or verify alphanumeric text strings** using **trained character models**.



Read Text (OCRMax)

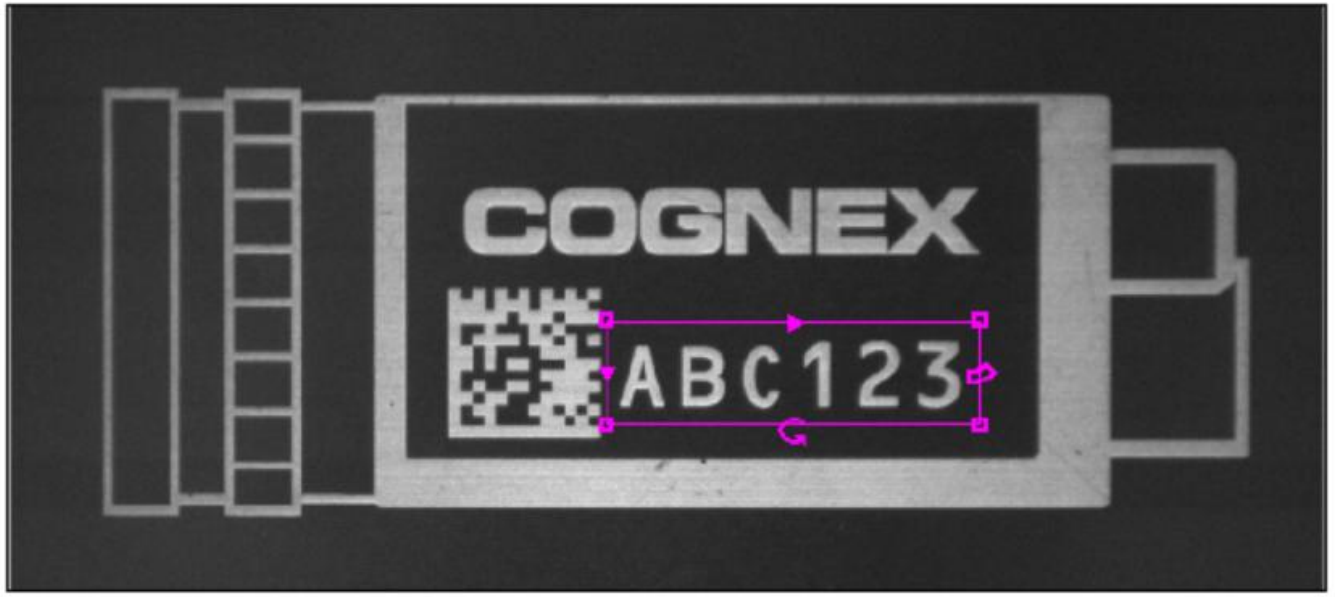
- **Optical Character Recognition (OCR)** is the process of reading characters by comparing detected character patterns with **trained character models**. The recognized characters are then combined and output as a **text string**.
- During the recognition process, the tool compares each detected character with the **trained characters in its database** to determine the **best match**.

Read Text (OCRMax)

- **Optical Character Verification (OCV)** is the process of reading characters and comparing the detected patterns with **trained character models** to verify that the **expected text string** is present.
- During verification, the tool compares each detected character **only with the trained character model corresponding to its expected identity** to determine whether it is a valid match.

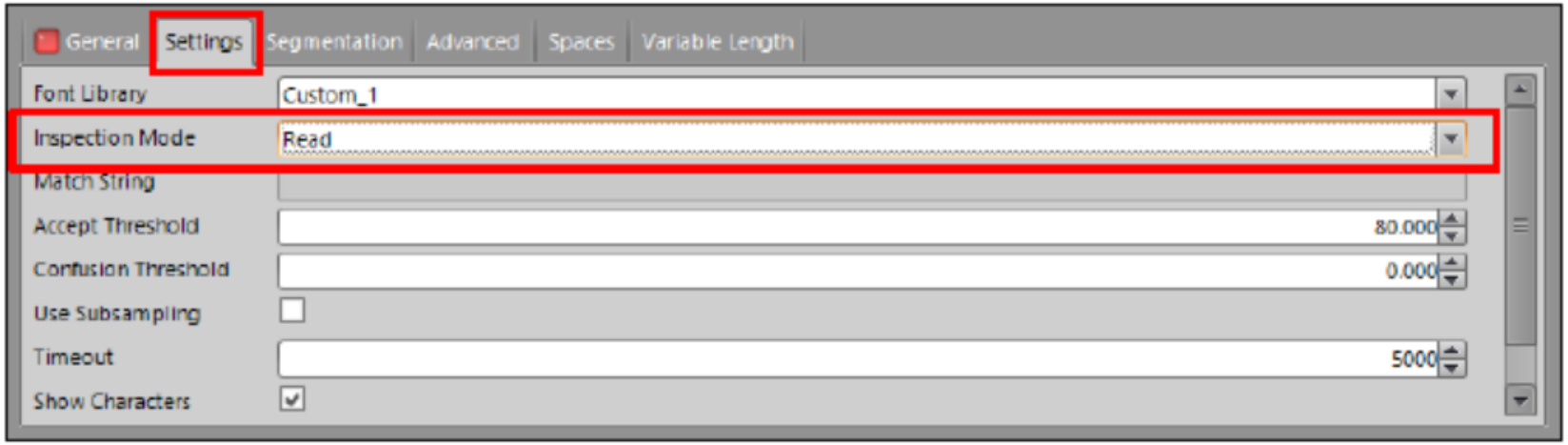
Read Text (OCRMax)

- Position the **region** over the text string you want to read or verify. The line of text within the region is divided into **individual character images**, with each character enclosed in a **non-editable rectangle**. The region defines the approximate **location, angle, and skew** of the line of text.

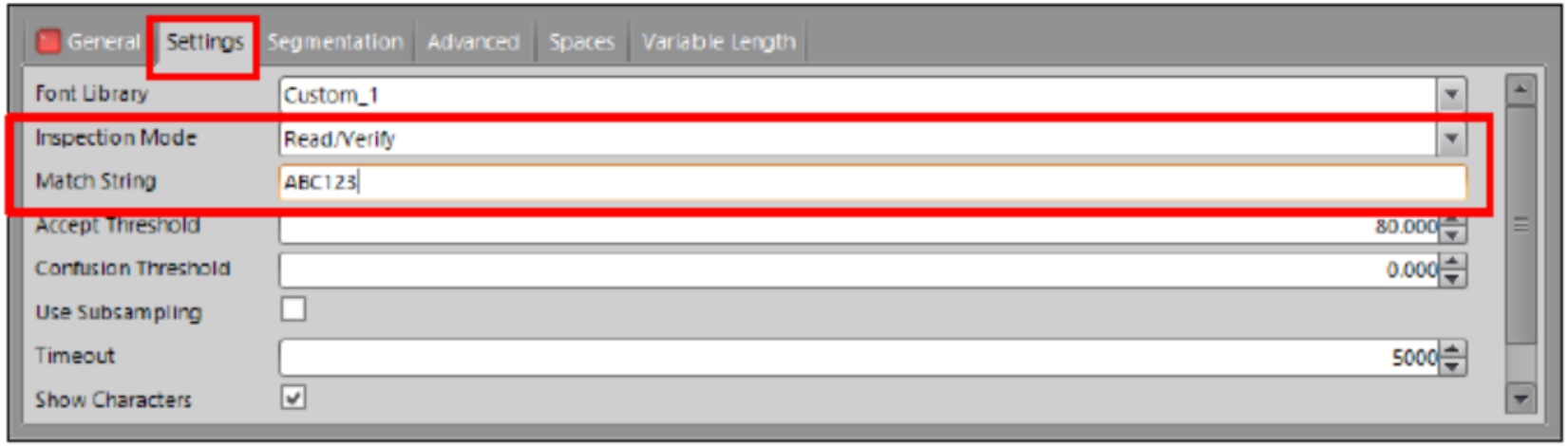


Read Text (OCRMax)

Read



Read/Verify



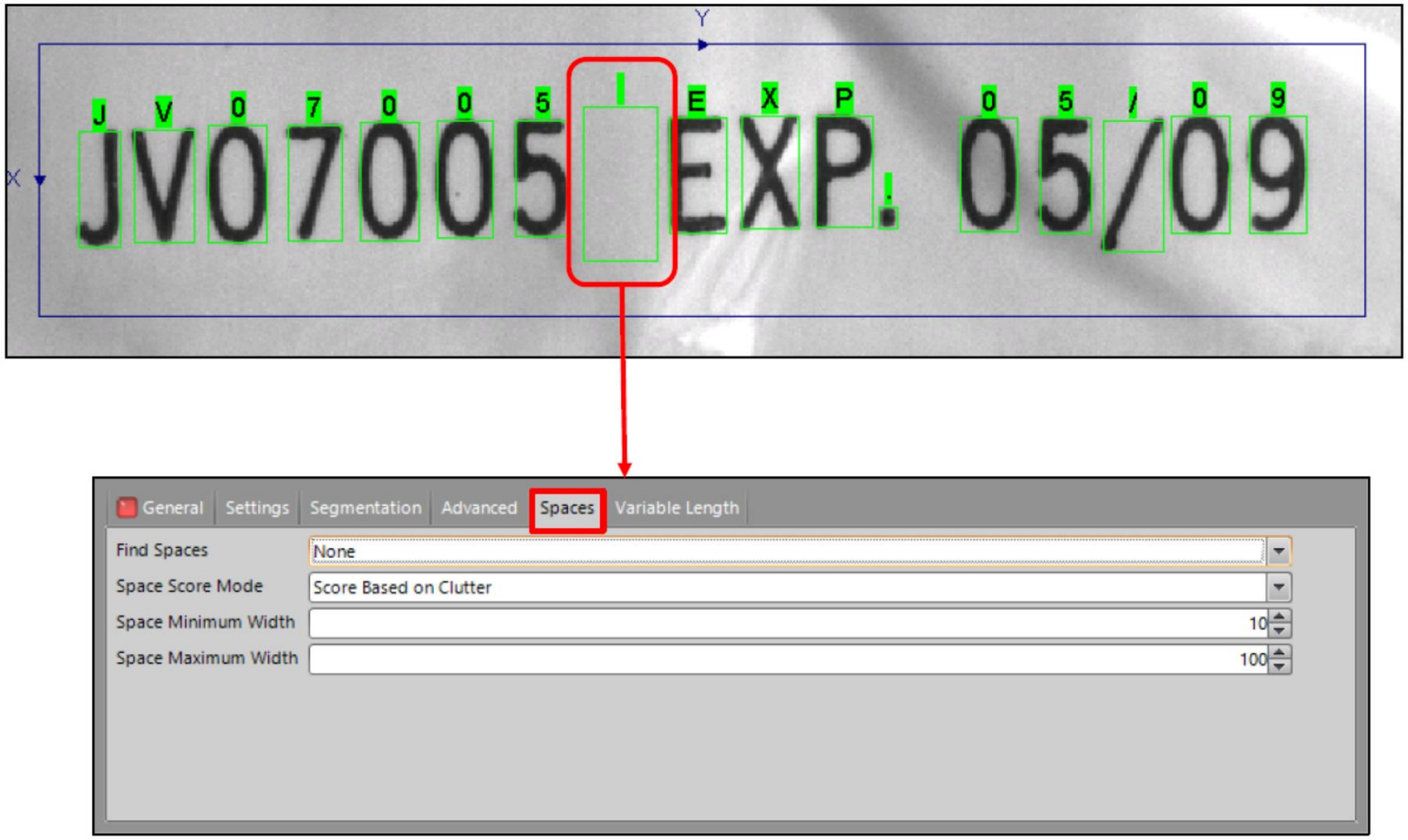
Read Text (OCRMax)

The **Inspection Mode** specifies how the function operates during **run-time**.

- **Read** is the default mode. The function attempts to **read the characters** in the ROI based on the trained character models.
- **Read/Verify** – The function first attempts to **read the characters** in the ROI and then verifies that the decoded text matches the **text string specified in the Match String field**.
- **Match String** – Specifies the **expected text string** that must be correctly matched when the **Inspection Mode** is set to **Read/Verify**.

Read Text (OCRMax)

Spaces Tab

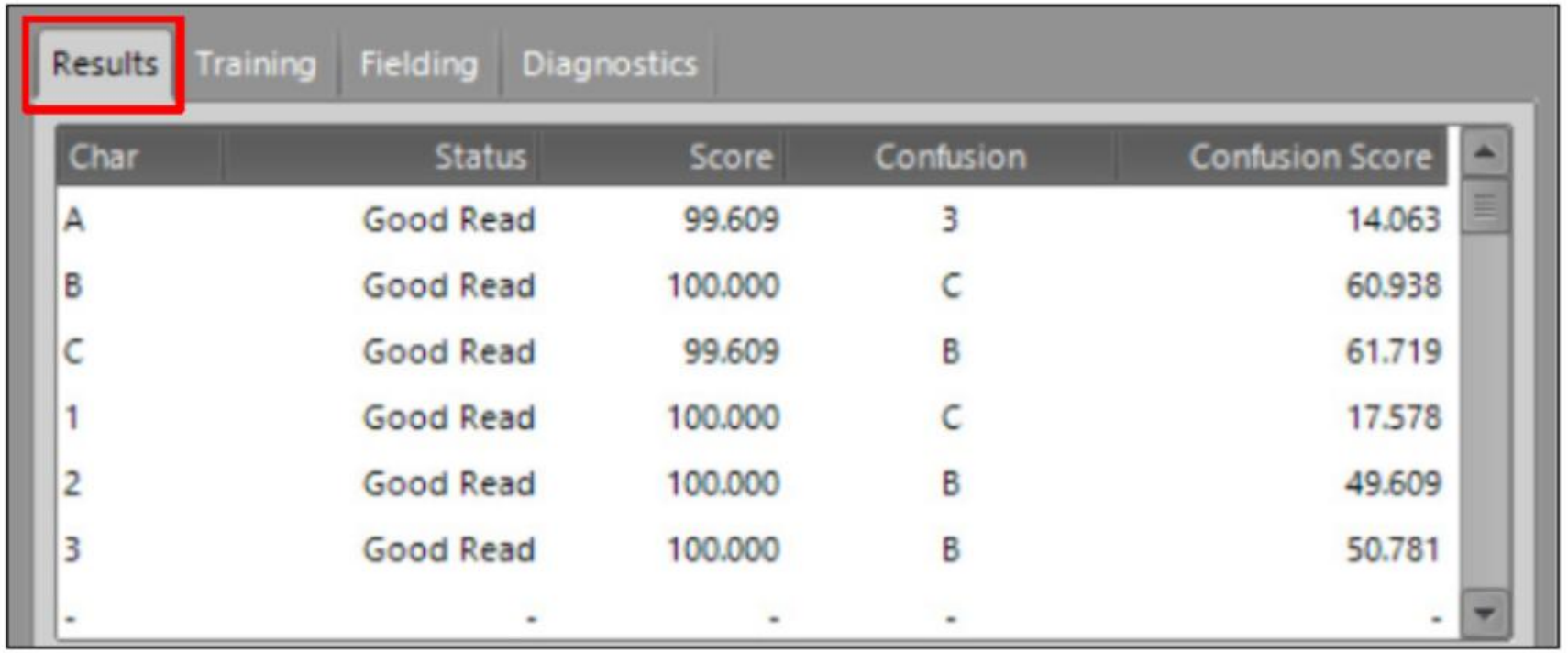


Read Text (OCRMax)

- The **Spaces** tab specifies how the function handles **spaces between characters**, when necessary. The OCRMax function evaluates **inter-character gaps** and can classify them as space characters based on the selected settings.
- **Find Spaces** – Specifies how the function inserts **space characters** into gaps between characters.
  - **None** is the default – The function **never inserts a space character**, regardless of the size of the gap between characters.
  - **Insert Single Space** – The function inserts **one space character** for each inter-character gap, regardless of the size of the gap.
  - **Insert Multiple Spaces** – The function inserts **multiple space characters** for each inter-character gap, based on the specified number.

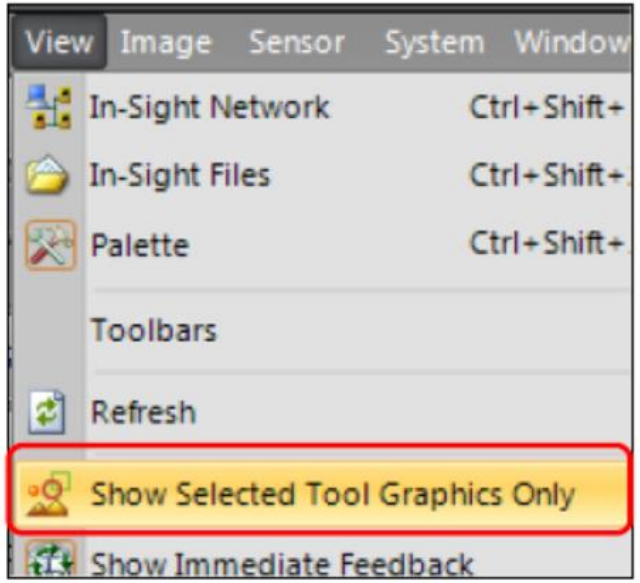
Read Text (OCRMax)

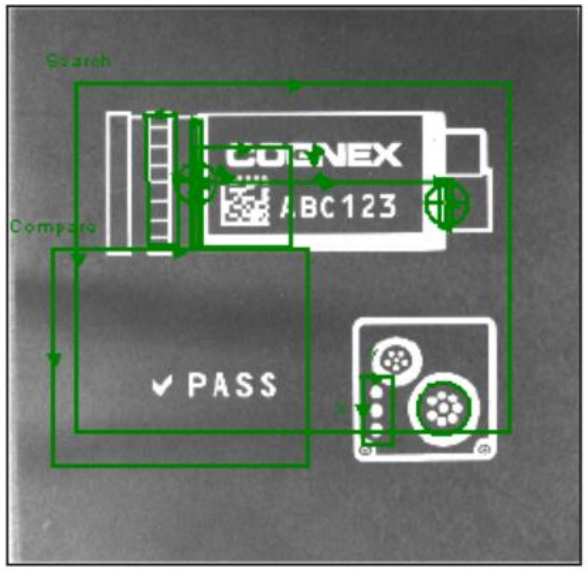
- Results Tab

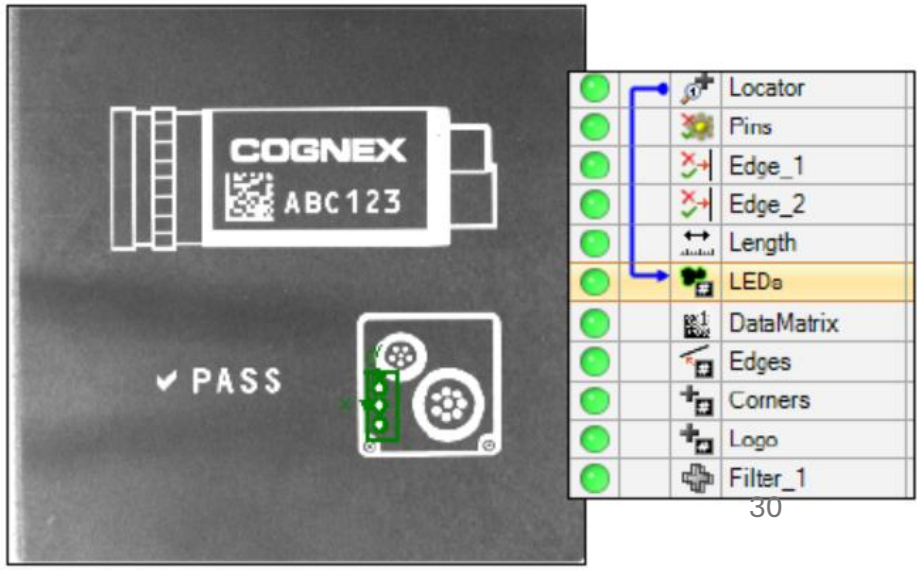


Char	Status	Score	Confusion	Confusion Score
A	Good Read	99.609	3	14.063
B	Good Read	100.000	C	60.938
C	Good Read	99.609	B	61.719
1	Good Read	100.000	C	17.578
2	Good Read	100.000	B	49.609
3	Good Read	100.000	B	50.781
-	-	-	-	-

Show Selected Tool Graphics Only







- When many tools are listed in the **Results** list, it can be difficult to determine which tool corresponds to which region in the image. By default, **all tool regions are displayed**.
- You can display the **region associated with the selected tool** to identify its location on the image.

Image Filter Tools

Image Filter Tools

- The **Image Filter Tools** are used to **enhance an image or a specific area of an image** for further image analysis.

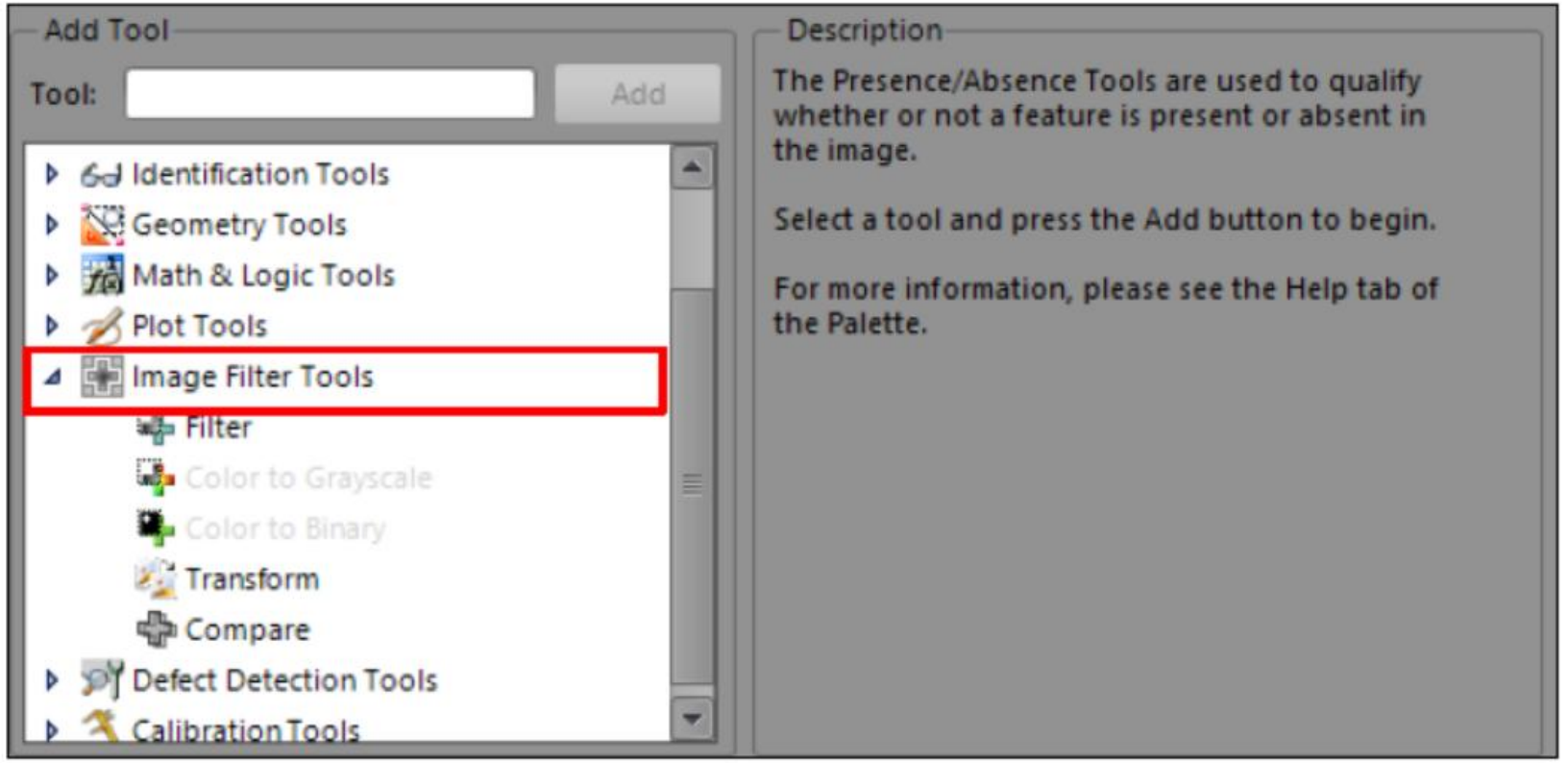


Image Filter Tools

- The **Filter Tool** applies **local and global image-enhancement techniques** to a region of an image. The **Filter Type** determines the enhancement method, and the tool outputs a **Tool Image** that can be referenced by other **Location or Inspection Tools**.
- The Filter Tool is a **multipurpose image-enhancement tool** that produces a **color or grayscale output image** to improve low-contrast images or reveal features that may be difficult to detect.
- The tool helps **accentuate desired features and minimize unwanted features**, making inspection applications more **reliable and repeatable**.

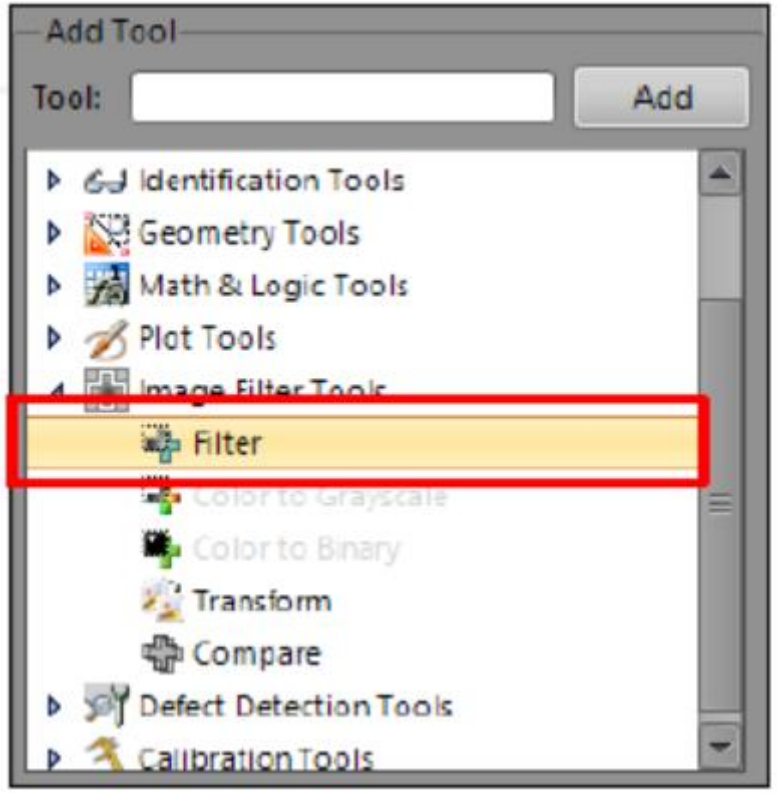
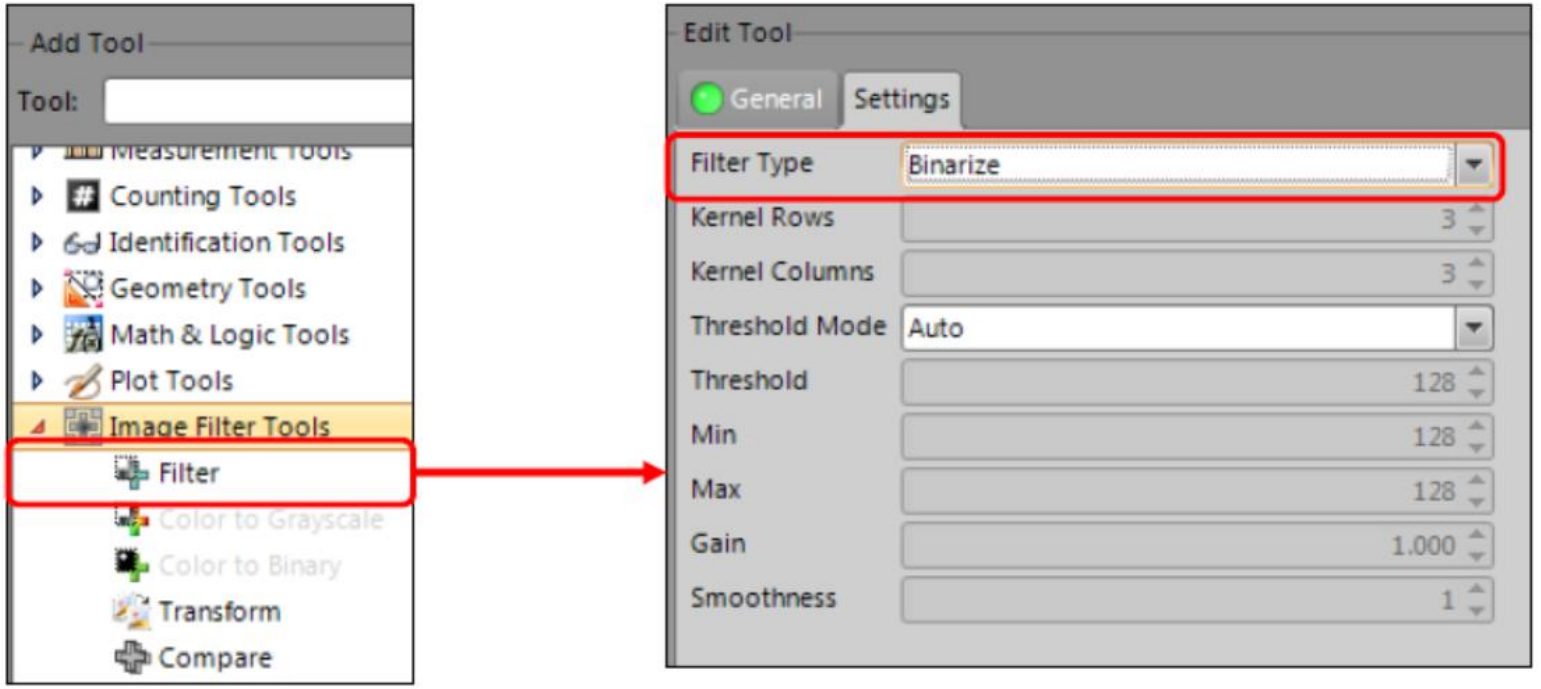


Image Filter Tools



Binarize



Edge Magnitude

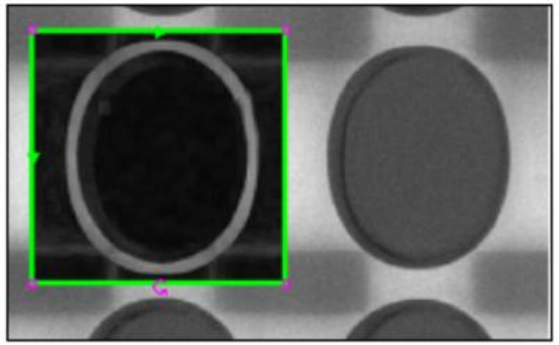


Image Filter Tools

The **Filter Tool's Settings Tab** contains the following information:

- **Filter Type** – Defines the **image-enhancement technique** applied to the region. The enhanced image is automatically updated within the region.
- **Binarize** (default) – Converts the image to **black (0) or white (255)** based on a grayscale threshold. Pixel values **equal to or above the threshold** become white, while values **below the threshold** become black.
- **Clip** – Filters the image by comparing each pixel's **grayscale value** with specified minimum and maximum values. Pixel values outside the specified range are adjusted to the corresponding limits, helping make image features **more uniform**.

Image Filter Tools

- **Greyscale Distance** – Calculates the output pixel value as the **difference between the input pixel value and the threshold value**. For example, if the input pixel value is 100 and the threshold is 25, the output pixel value is **75**.
- **Invert** – Inverts the grayscale values of the image, producing a **negative** of the original image. The inverted value is calculated as: **Inverted Value = 255 – Original Value.**
- **Threshold Range** – Converts the image to **black and white** by comparing each input pixel with specified **minimum and maximum threshold values**. Pixel values **outside the threshold range** are converted to white, while values **within the range** are converted to black.

Image Filter Tools

- **Optical Density –** Filters the image by applying a **logarithmic inversion of pixel values** to represent the density of objects and features based on the amount of light passing through them. **Denser objects and features appear as lighter pixels** in the output image.

$$OD(x,y) = -\log\left(\frac{I(x,y)}{I_0}\right)$$

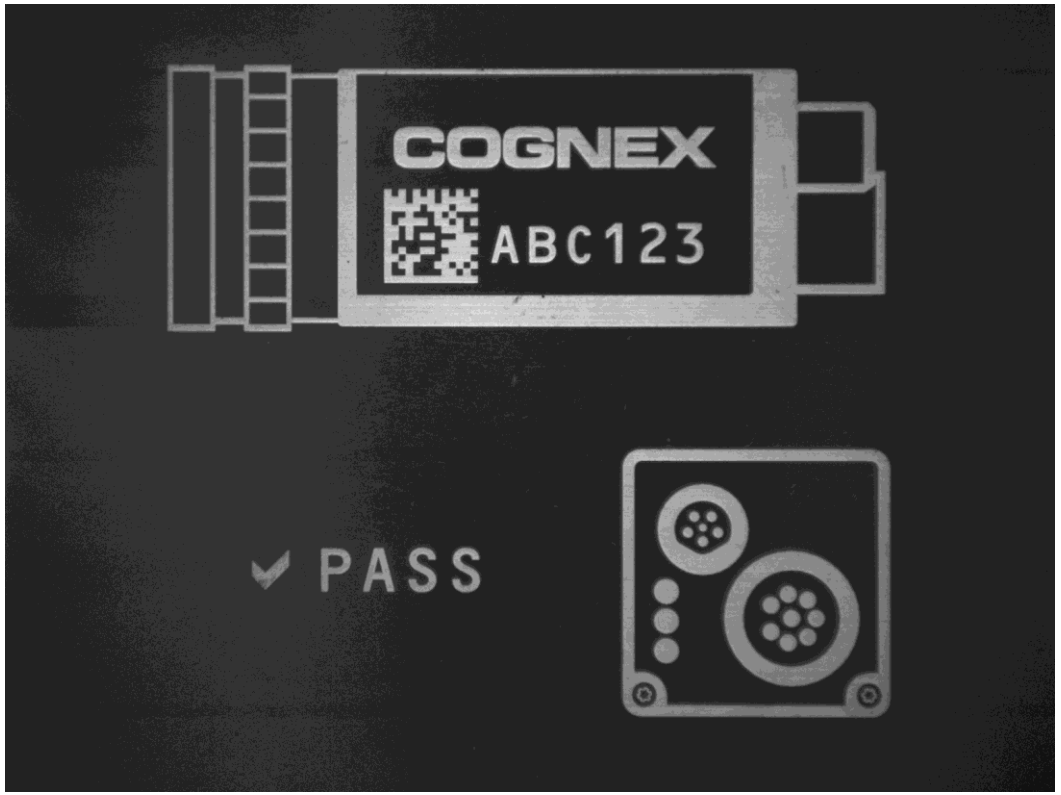
$I(x,y)$ = pixel intensity (light transmitted through at that point)

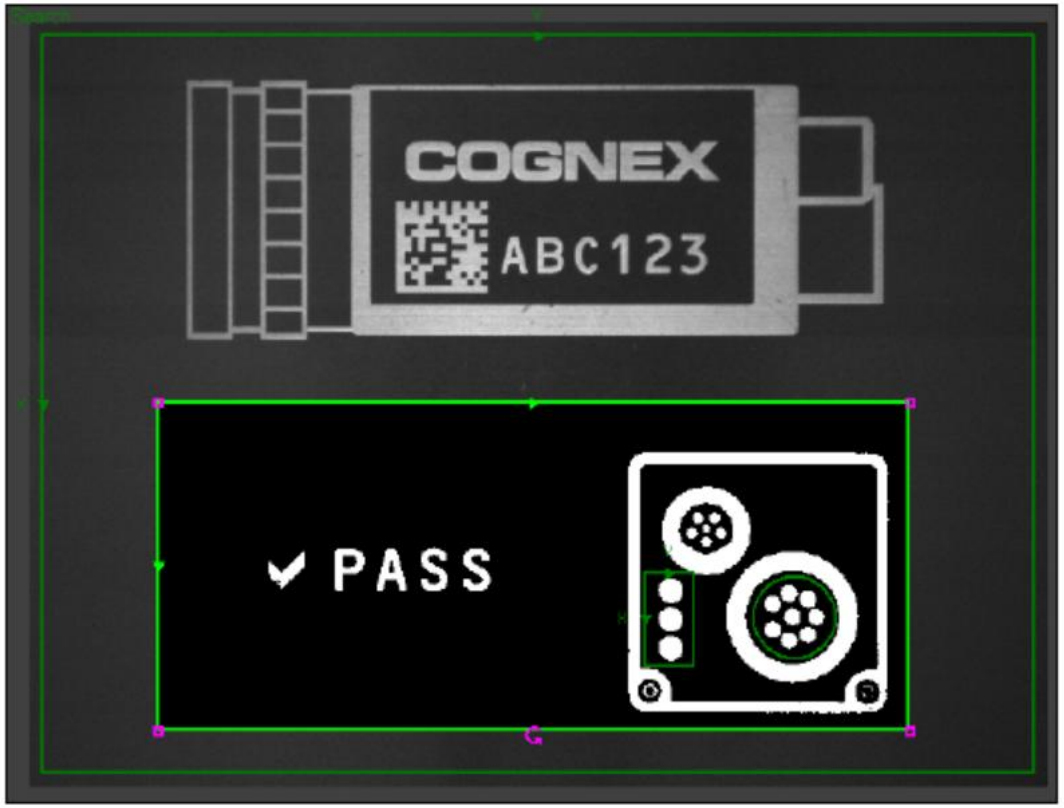
I_0 = reference maximum intensity (light with no object blocking)

Image Filter Tools

- Imagine shining light through a piece of plastic with a **thick part** and a **thin part**:
- In the raw grayscale image:
  - Thin region lets more light through → bright pixel.
  - Thick region blocks more light → dark pixel.
- In the **Optical Density image**:
  - Thick region (dark in original) → high OD → becomes bright.
  - Thin region (bright in original) → low OD → becomes dark.
- After OD filtering, **denser features show up lighter**, which is more intuitive for inspection (like how bones look bright in an x-ray).

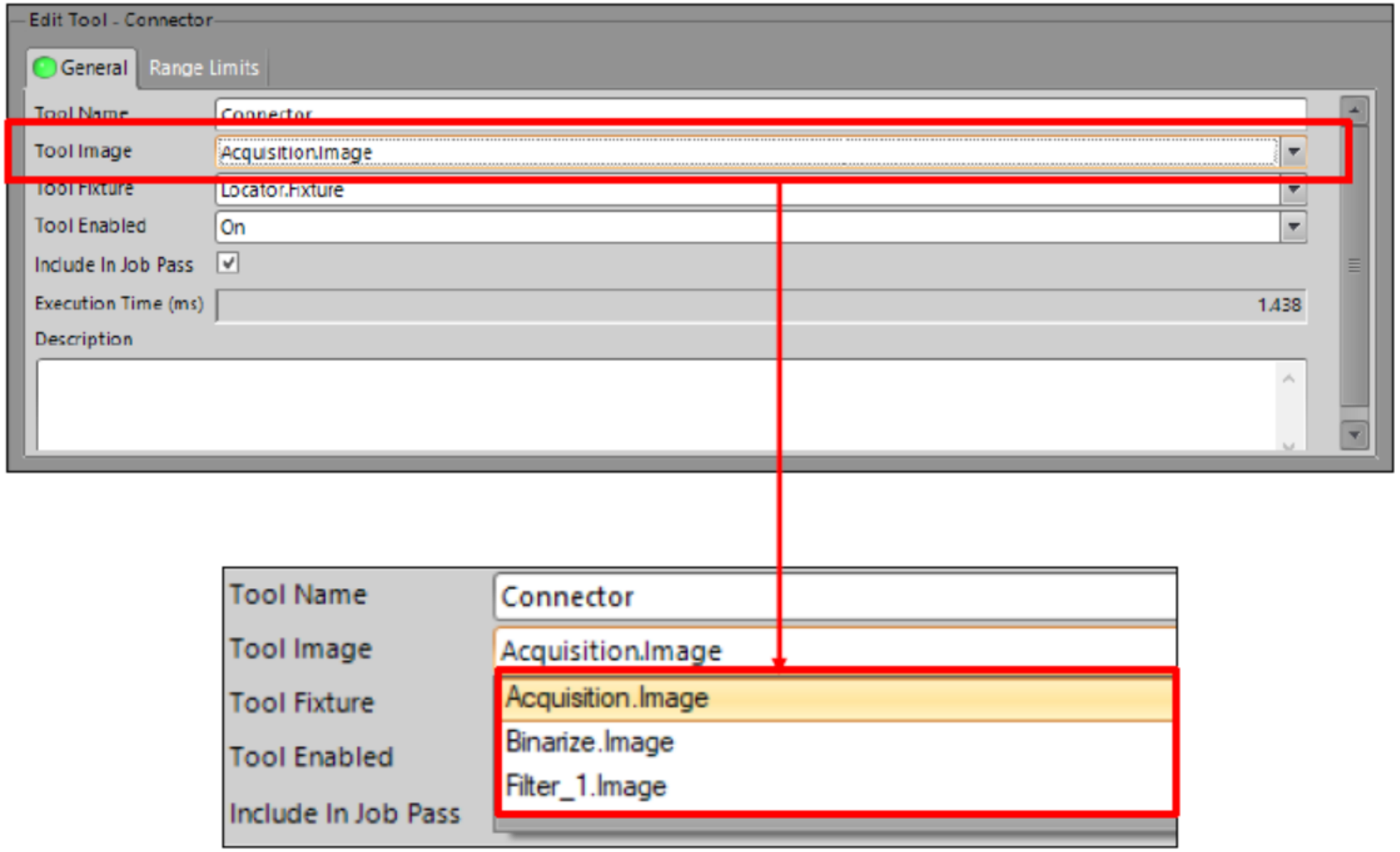
Binarize Filter Result





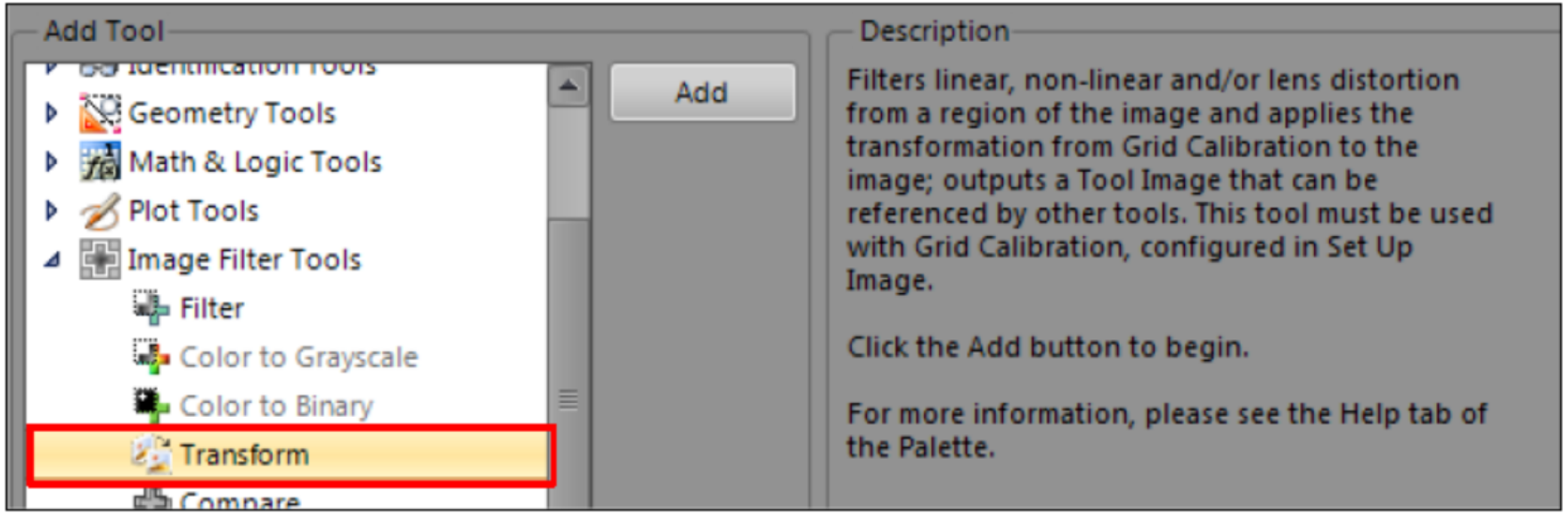
Binarize Filter Result

- Once you have created a filtered image, **EasyBuilder** allows you to select it for use with any **Vision Tool**.

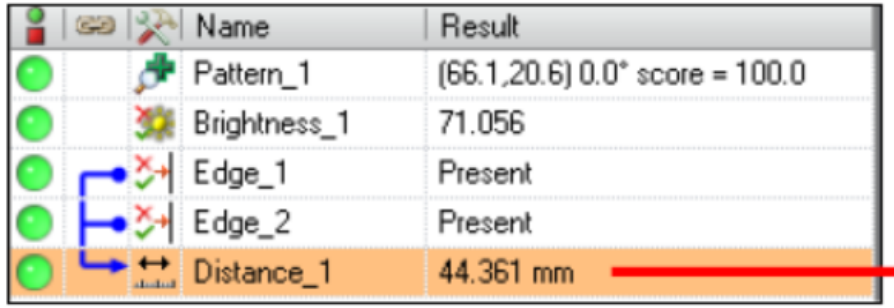


Transform Filter

- The **Grid Calibration** process stores conversion data for each distance (point-to-point or square-to-square) in the job. This information can then be applied to the original image to produce a new **Transform** image that is corrected for lens distortion.



Transform Filter



Results are based on original image, which may have distortion from lens

To decrease distortion:

1. Use Grid Calibration
2. Add **Transform** tool
3. Have measurement tool reference **Transform** in General tab:

